

ECSP3000 Project Brief

Designing and Implementing an Automated & Reproducible Testbed for Linux Rootkit Detectors, Aiming to Identify and Implement a Novel Rootkit/Detector Technique

Elliott Jelbert

35949244 - ej1g24@soton.ac.uk

Supervisor: Dr. BooJoong Kang

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1. Project Aims

Current research into the evaluation of rootkits and rootkit detectors rely on bespoke evaluation methods which cannot be robustly reproduced by third parties.

This tool aims to provide developers, researchers and sys-admins with an automated and reproducible testing framework for rootkits & rootkit detectors on the modern Linux kernel to foster future development and testing.

As an extension, I will use this tool to identify and evaluate current avenues in the modern (7.2) Linux Kernel for rootkit persistence and design, finishing with an evaluation using the aforementioned benchmarking tool.

2. Project Objectives

This project will have the following aims which will be used to gauge success:

1. To have a testing framework which, upon specifying a set of rootkits & rootkit detectors, provides timely & accurate results which can then be reproduced exactly on rerun.
2. The testing framework allows developers to easily input their own rootkit detectors and new rootkit PoCs to allow for future testing.

3. (Stretch) To research and develop a rootkit PoC for the modern Linux Kernel, following with an evaluation using this testing framework.
4. Using a dataset of results produced by the framework or, if successful, the aforementioned PoC, evaluate the current landscape of rootkit security on the modern Linux Kernel.

3. Limitations & Scope

In order to scope my project appropriately, here are a list of things that I currently plan on **not** doing.

1. My testing framework and therefore my rootkit research will only be in regards to Kernel Space Rootkits.
2. All detector techniques compatible with the tool must run **on host**. This rules out any hypervisor techniques or off-host memory inspection as detection techniques.
3. Testing rootkits that implement anti-sandbox measures fall out of scope for this project.